

MULTI-AGENT DATA ANALYSIS AND INSIGHT GENERATION PLATFORM FOR FINANCIAL AND SALES DATA

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Motivation

- Despite the availability of data, many organizations lack the technical expertise required to analyze and interpret it effectively.
- Traditional data analysis tools often require statistical knowledge, programming skills, or manual report interpretation.
- Existing AI-based systems using LLMs may generate unverified or hallucinated insights, leading to unreliable conclusions.

Introduction

- The project is being implemented as a state-based data analysis pipeline for sales and financial CSV datasets.
- Four pipeline stages are currently connected: structural profiling, semantic tagging, preprocessing, and statistical analysis.
- The system transforms raw records into a cleaned dataset, a preprocessing audit trail, quality metrics, statistics, and chart files.
- Mathematical operations are performed with Python libraries; the language model is used for semantic interpretation and processing rules.
- The current implementation provides a working backend foundation for the planned user-facing platform.

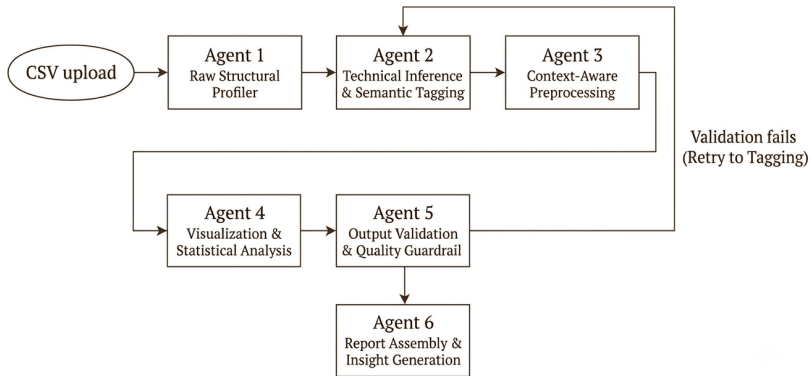
Objectives

- Develop an autonomous multi-agent system that converts raw sales and financial data into reproducible analytical outputs.
- Implement semantic, context-aware preprocessing instead of applying the same cleaning rule to every column.
- Evaluate the pipeline using representative datasets, focused unit tests, and generated diagnostics.
- Establish a reliable backend foundation before completing report generation and the user interface.

Scope of Project

- Current implementation: sales and financial datasets in CSV format
- Completed: structural profiling, semantic tagging, preprocessing, and pipeline routing
- Completed: data quality scoring, audit logging, derived business metrics, and chart generation
- In progress: richer statistical analysis, validation guardrails, and report assembly
- Future scope: web interface and support for Excel, JSON, and Parquet inputs

Methodology – Current 4-Stage Pipeline



Agent 1 – Raw Structural Profiler

- Reads the input CSV and records shape, dtypes, missingness, uniqueness, and duplicates.
- Produces a raw structural profile without changing the source data.
- Caches the DataFrame for the downstream stages.

Agent 2 – Technical Inference & Semantic Tagging

- Sniffs numeric, datetime, boolean, and string types using local heuristics.
- Uses Groq-based semantic tagging when available and falls back to deterministic rules when it is not.
- Creates a per-column blueprint containing type, semantic tag, scaling, imputation, and analysis policies.
- Passes the blueprint to preprocessing instead of hard-coding one rule for every column.

Agent 3 – Context-Aware Preprocessing

- Coerces columns according to the semantic blueprint and cleans international currency formats.
- Standardizes text and null variants while preserving identifiers.
- Removes duplicates, applies configured imputation, and clips permitted outliers with IQR bounds.
- Extracts date features and derives metrics such as profit, margin, and revenue per unit.
- Records scaling parameters, an audit log, and a 0–100 data quality score.

Agent 4 – Visualization & Statistical Analysis

- Calculates descriptive statistics, correlations, distributions, growth, and seasonality.
- Detects anomalies using a configurable z-score threshold.
- Computes regression trends and category rankings when the dataset supports them.
- Saves reproducible PNG charts and structured statistics to the pipeline state.

Planned Extension – Validation Guardrail

- Cross-check statistical outputs against the cleaned dataset.
- Verify that generated charts match the corresponding statistics.
- Check whether every future insight is supported by measurable evidence.
- Stop or flag the pipeline when validation rules fail.
- This stage is planned and is not included in the current four-stage run.

Planned Extension – Report Assembly

- Read validated statistical results from the pipeline state.
- Generate human-readable explanations with traceable evidence.
- Compile charts, quality metrics, and findings into the final report.
- Add an interface for dataset upload and progress monitoring.
- This stage remains part of the next implementation phase.

Current Progress – Verified Run

- **Input tested:** Amazon sales CSV with 10,000 rows and 21 columns.
- **Agent 1:** 210,000 cells profiled; 0 missing cells and 0 duplicate rows in the run.
- **Agent 2:** All 21 columns received a semantic schema blueprint and processing policy.
- **Agent 3:** Cleaned CSV, scaling parameters, column ledger, preprocessing log, and quality metrics generated.
- **Agent 4:** Statistics and chart infrastructure executed through the LangGraph pipeline.

Generated Outputs

- Cleaned dataset exported to `backend/outputs/cleaned_data.csv`
- Run metadata exported to `agent_run_diagnostics.json`
- Correlation heatmap and numeric distribution charts generated as PNG files
- Category and payment-method distribution charts generated from the processed data
- Reliability metadata records stage confidence, evidence, and decision readiness

Testing and Engineering Progress

- Focused tests cover semantic-tagging chunking, currency parsing, duplicate consistency, chart overwrite behavior, and pipeline diagnostics.
- Chart generation clears stale PNG files before each run to avoid reporting outdated results.
- Pipeline errors are accumulated in shared state and conditional routing stops downstream stages when a critical stage fails.
- The current backend is reproducible from a defined input path and writes structured diagnostics for inspection.

Current Analysis

- **Deterministic analysis:** Pandas, NumPy, SciPy, and Matplotlib perform the numeric calculations and visualizations.
- **Semantic flexibility:** The schema blueprint lets preprocessing respond to column meaning while retaining a fallback path when the LLM is unavailable.
- **Traceability:** The pipeline preserves an audit log, scaling parameters, quality metrics, charts, and diagnostics for each run.
- **Current boundary:** The backend is functional, but the final validation agent, natural-language report assembly, and user interface are not complete yet.

Mid-Progress Timeline

- **Completed:** project architecture, GraphState, Agents 1–3, conditional pipeline routing, preprocessing audit trail, quality scoring, and focused tests.
- **Current:** Agent 4 statistical analysis and chart generation, verified on the Amazon sales dataset.
- **Next:** complete output validation and connect statistics to evidence-backed insight generation.
- **Final phase:** report assembly, upload workflow, frontend presentation, broader datasets, and end-to-end evaluation.

Remaining Work

Work package	Target outcome
Validation guardrail	Check statistics, charts, and evidence
Insight generation	Produce traceable natural-language findings
Report assembly	Combine outputs into a user-facing report
Interface	Upload data and display progress/results
Evaluation	Benchmark accuracy, robustness, and run-time

Conclusion

- **Backend foundation achieved:** A working LangGraph pipeline now carries a CSV dataset through four specialized stages.
- **Core transformation achieved:** Profiling, semantic tagging, context-aware preprocessing, quality scoring, and statistical chart generation are operational.
- **Evidence available:** The verified run produces cleaned data, diagnostics, audit records, statistics, and chart artifacts.
- **Next objective:** complete validation, evidence-backed report generation, interface integration, and broader evaluation.

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